

Efstratios Tsoukanis

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POSITIONS

Visiting Scholar Current
Institute of Mathematical Sciences, Claremont Graduate University, Claremont, CA

- Research with Distinguished Research Professor Hrushikesh Mhaskar on computational harmonic analysis and machine learning, supported by the US Office of Naval Research.

Research Assistant Professor 2024–2025
Institute of Mathematical Sciences, Claremont Graduate University, Claremont, CA

EDUCATION

Ph.D. in Mathematics 2024
University of Maryland, College Park

- Dissertation: *G-Invariant Representations using Coorbits*
- Advisor: Professor Radu Balan

M.S. in Mathematics 2019
University of Crete, Heraklion

- Thesis: *Fourier multiplier and Fefferman counter-example*
- Advisor: Professor Themistoklis Mitsis

B.S. in Mathematics 2016
University of Crete, Heraklion

- Erasmus exchange at the University of Barcelona, 2014

RESEARCH INTERESTS

Applied harmonic analysis and geometric machine learning: group-invariant representations (embeddings of orbit spaces, injectivity and bi-Lipschitz properties), phase retrieval, kernel methods and approximation theory.

PUBLICATIONS

Journal articles

- R. Balan and **E. Tsoukanis**. G-invariant representations using coorbits: Bi-Lipschitz properties. *Advances in Computational Mathematics* 52(5), 92, 2026. arXiv:2308.11784
- N. Dym, M. Wellershoff, **E. Tsoukanis**, D. Levy, and R. Balan. Quantitative bounds for sorting-based permutation-invariant embeddings. *IEEE Transactions on Information Theory*, to appear, 2026. arXiv:2510.22186
- H. N. Mhaskar, **E. Tsoukanis**, and A. D. Jagtap. An approximation theory perspective on machine learning. *Neural Networks*, 108841, 2026. ScienceDirect

Conference proceedings

- H. N. Mhaskar, R. O'Dowd, and **E. Tsoukanis**. Active learning classification from a signal separation perspective. *International Conference on Sampling Theory and Applications (SampTA)*, Vienna, Austria, 2025. arXiv:2502.16425

5. R. Balan and **E. Tsoukanis**. Relationships between the phase retrieval problem and permutation invariant embeddings. *International Conference on Sampling Theory and Applications (SampTA)*, New Haven, CT, 2023. arXiv:2306.13111

Preprints

6. R. Balan, **E. Tsoukanis**, and M. Wellershoff. Stability of sorting based embeddings. arXiv:2410.05446, 2024.
7. R. Balan and **E. Tsoukanis**. G-invariant representations using coorbits: Injectivity properties. arXiv:2310.16365, 2023.

TALKS

- 2025 *Active learning classification from a signal separation perspective*. SampTA 2025, Vienna, Austria.
- 2023 *Relationships between the phase retrieval problem and permutation invariant embeddings*. SampTA 2023, Yale University, New Haven, CT.
- 2023 *Stability properties of coorbit embeddings*. CodEx Seminar.
- 2022 *Noise stability of functions with low influences*. Computational Learning Theory and Fourier Analysis Summer School.

TEACHING

Teaching Assistant, University of Maryland Calculus II and III, Linear Algebra, Differential Equations	2018–2023
Teaching Assistant, University of Crete Calculus I, Analysis I	2015–2016

SERVICE AND OTHER ACTIVITIES

- Referee for *Foundations of Computational Mathematics* and *Applied and Computational Harmonic Analysis*
- Research Interaction Team in Applied Harmonic Analysis: Fall 2021, Fall 2023
- Research Interaction Team in Deep Learning: Fall 2022, Spring 2023, Fall 2023

SKILLS

Programming: Python, MATLAB, $\text{\LaTeX}$ **Languages:** Greek, English